

Experimental Demonstration of 3D Optical Wireless Positioning with Single Beam-Steered LED and Empirical Radiation Pattern

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Abstract—TBD

Index Terms—Optical wireless positioning, visible light positioning, beam steering, 3D localization, radiation pattern modeling.

I. INTRODUCTION

II. WORKING PRINCIPLES OF 3D OWP

A. System Model

B. Direction Finding Formulation

C. Distance Recovery: Active vs. Passive Receiver Approaches

This subsection explores two distinct methods for distance recovery: (a) *Active Receiver Orientation*: Assuming the receiver possesses reorientation capabilities (emulated via the robotic arm's end-effector), the PD is actively aligned to face the LED. This establishes a direct line-of-sight link to recover the distance, requiring a total of $K + 1$ orientation steps. (b) *Passive Fixed Receiver*: The PD remains strictly static, and the distance is mathematically derived solely from the Received Signal Strength (RSS) information gathered during the K initial beam-steering orientations of the transmitter.

D. Empirical Radiation Modeling for Estimation Enhancement

This subsection addresses the limitations of classical optical channel models: (a) It presents a theoretical discussion contrasting the ideal Lambertian irradiance model with the empirical, quasi-Lambertian radiation pattern exhibited by Commercial Off-The-Shelf (COTS) LEDs. (b) It proposes a novel mathematical framework to integrate this high-fidelity empirical radiation data into the estimation algorithms (e.g., via Look-Up Tables or non-linear calibration functions) to mitigate spatial biases and enhance 3D positioning accuracy.

III. EXPERIMENTAL FRAMEWORK AND IMPLEMENTATION DETAILS

A. 3D Robotic Testbed and Hardware Upgrades

To scale the evaluation beyond preliminary conference results, several hardware upgrades were implemented:

- **Thermal Management and Nominal Current Operation:** An aluminum heat sink was integrated into the

motorized gimbal, and the transmitter LED was mounted on a star PCB. This thermal dissipation enhancement allowed the LED to operate safely at its nominal forward current of 1A (a significant increase from the 750 mA used in our previous work), thereby improving the Signal-to-Noise Ratio (SNR) and reducing potential thermal implications.

- **Receiver FOV and High-Density Spatial Grid:** To thoroughly validate the spatial robustness of the SBL method, the experimental evaluation grid will be massively densified from the baseline of 300 positions. Maintaining a constant spatial resolution of 0.2 m, two receiver configurations could be considered:

- 1) *Baseline PD (BPX 61)*: Constrained by a narrower half-angle FOV ($\approx 60^\circ$), the reliable evaluation area is limited to 2.4×2.4 m. Depending on the evaluated height range, this configuration yields 845 positions ($h \in [0.4, 1.4]$ m) or 1014 positions ($h \in [0.2, 1.4]$ m).
- 2) *Wide-Angle PD Upgrade (S6967)*: By transitioning to a receiver with a wider half-angle FOV ($\approx 90^\circ$), the testbed can be expanded to its physical limit of 3.0×3.0 m. This allows for a massive dataset collection of 1280 positions ($h \in [0.4, 1.4]$ m) or 1536 positions ($h \in [0.2, 1.4]$ m).

In all considered scenarios, the dataset scale is increased to provide a comprehensive statistical evaluation of the peripheral error degradation.

B. Kinematics and Coordinate Transformations

This subsection details the rigid-body transformations required for accurate ground-truth mapping: (a) *Transmitter Kinematics*: Derivation of the rotation matrices and coordinate transformations governing the two-axis motorized gimbal mechanism. (b) *Receiver Kinematics*: Formulation of the equations mapping the desired PD spatial coordinates within the global testbed reference frame to the local position and orientation vectors of the robotic arm's coordinate system.

C. Experimental Protocol for SBL Data Acquisition

In addition to spatial variations, this protocol includes a specific procedure to validate the system's robustness against receiver misalignment. Measurements were recorded at identical spatial positions while systematically varying the PD's tilt (inclination and azimuth). This aims to experimentally demonstrate the tilt-agnostic properties of the ratio-based SBL method, particularly concerning direction-finding performance.

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D. Experimental Protocol for Empirical Radiation Mapping

Details the automated procedure utilized to acquire the high-resolution empirical radiation pattern of the transmitter LED, capturing dense angular RSS measurements to construct a faithful 3D spatial map of the optical beam.

IV. EXPERIMENTAL RESULTS AND DISCUSSION

A. Baseline Performance: Direction Finding and Positioning

B. System Robustness: The Impact of Receiver Tilt

C. Performance Enhancement via Empirical Radiation Modeling

This is the capstone results section. It compares the baseline estimation errors (which assume an ideal Lambertian source) against the improved estimation obtained by integrating the empirical radiation pattern. It demonstrates how compensating for the quasi-Lambertian asymmetries minimizes peripheral errors and significantly reduces both the direction-finding error (α) and the 3D positioning RMSE.

V. CONCLUSION AND FUTURE WORKS

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REFERENCES

- [1] K. Acuna-Condori *et al.*, “Robust visible light positioning: a comparative study of machine learning estimators in spatial segmentation and transmitter failure scenarios,” in *IEEE ICC*, 2025, pp. 3845–3851.
- [2] K. Acuna-Condori *et al.*, “Robust optical wireless positioning with beam steering: Pose estimation and tilt robustness,” 2026, accepted for presentation at IEEE International Conference on Communications (ICC), to appear, 2026.
- [3] K. Acuna-Condori *et al.*, “Model-based beam-steered optical wireless positioning with single-led single-photodiode for 3d localization,” 2025, manuscript submitted to IEEE Transactions on Communications.

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